

FATE: Monitoring DIC

2025 Fall Samples

2026-05-22

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```
##Setup - Change things here & write any notes
```

```
#identify section  
cat("Setup Information")
```

```
## Setup Information
```

```
##### Run information - PLEASE CHANGE  
Date_Run = "11/21/25" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = "Check standards not great on this run. Duplicates were out of range for a few dock samples"  
samples <- c("FATE") #whatever identifies your samples within the same names  
samples_pattern <- paste(samples, collapse = "|")  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_" #what did you name your check standards?  
crm_name = "CRM|crm" #what did you name your CRMS?  
  
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
raw_file_name = "Raw Data/FATE_DIC_202511.txt"  
  
#file path and name for raw all peaks file  
raw_allpeaks_name = "Raw Data/FATE_DIC_202511_allpeaks.txt"  
  
#file path and name of processed data file  
processed_file_name = "Processed Data/FATE_Processed_DIC_202511.csv"  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
  #Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"  
  
#qaqc log file path for this year  
Log_path = "Raw Data/FATE_DIC_QAQClog_2025.csv"
```

```
##Set Up Code
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:  
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 3  
##   sample_name      ic_raw run_datetime  
##   <chr>          <dbl> <chr>  
## 1 FATE_Dock_20251002_A 27.1 11/20/2025 11:01:48 PM
```

```
## 2 FATE_Dock_20251002_B 28.2 11/20/2025 11:29:28 PM
## 3 FATE_Flume_20251002_A 30.9 11/20/2025 11:52:35 PM
## 4 FATE_Flume_20251002_B 30.6 11/21/2025 12:15:49 AM
## 5 FATE_Dock_20251103_A 28.0 11/21/2025 12:39:05 AM
## 6 FATE_Dock_20251103_B 18.5 11/21/2025 1:00:02 AM
```

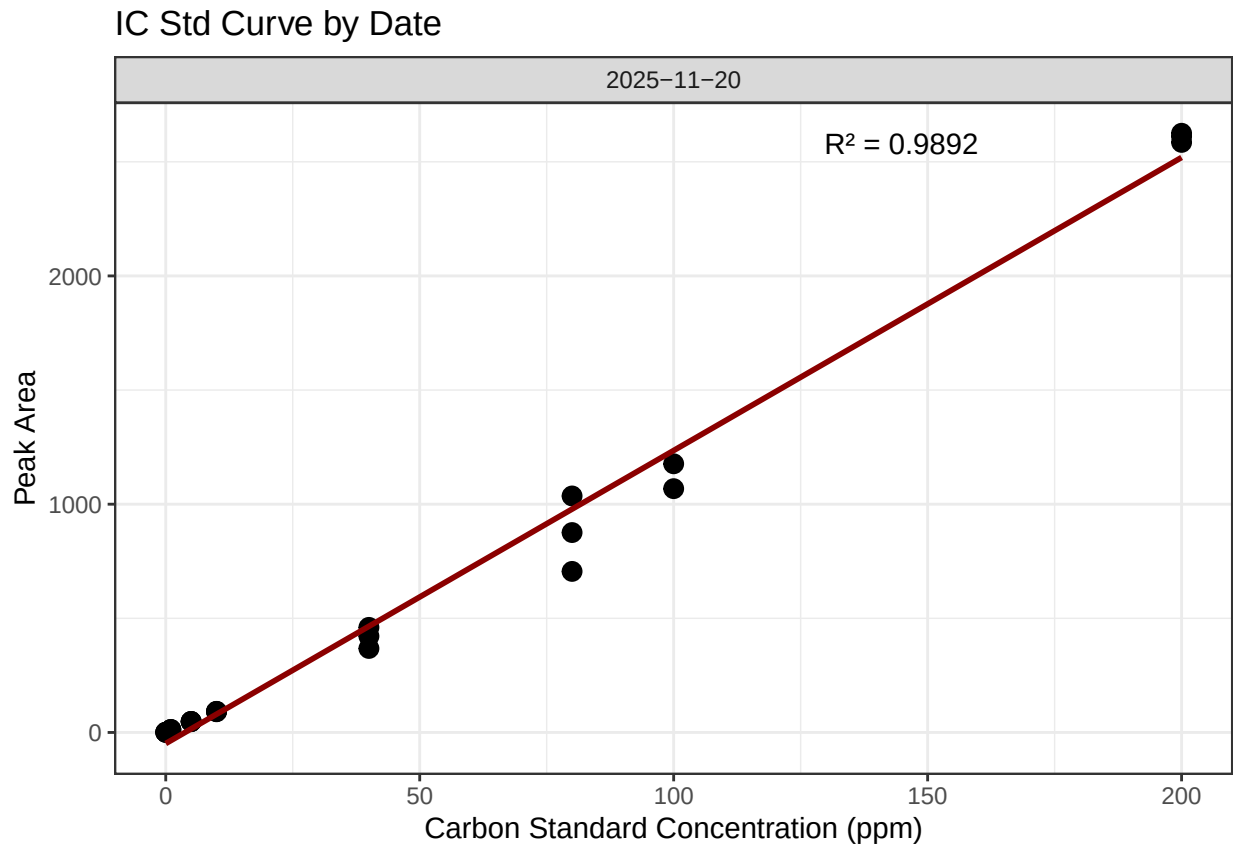
0.3 Assessing Standard Curves

```
## Assess the Standard Curves
```

```
## New names:
## * ' ' -> '...18'
```

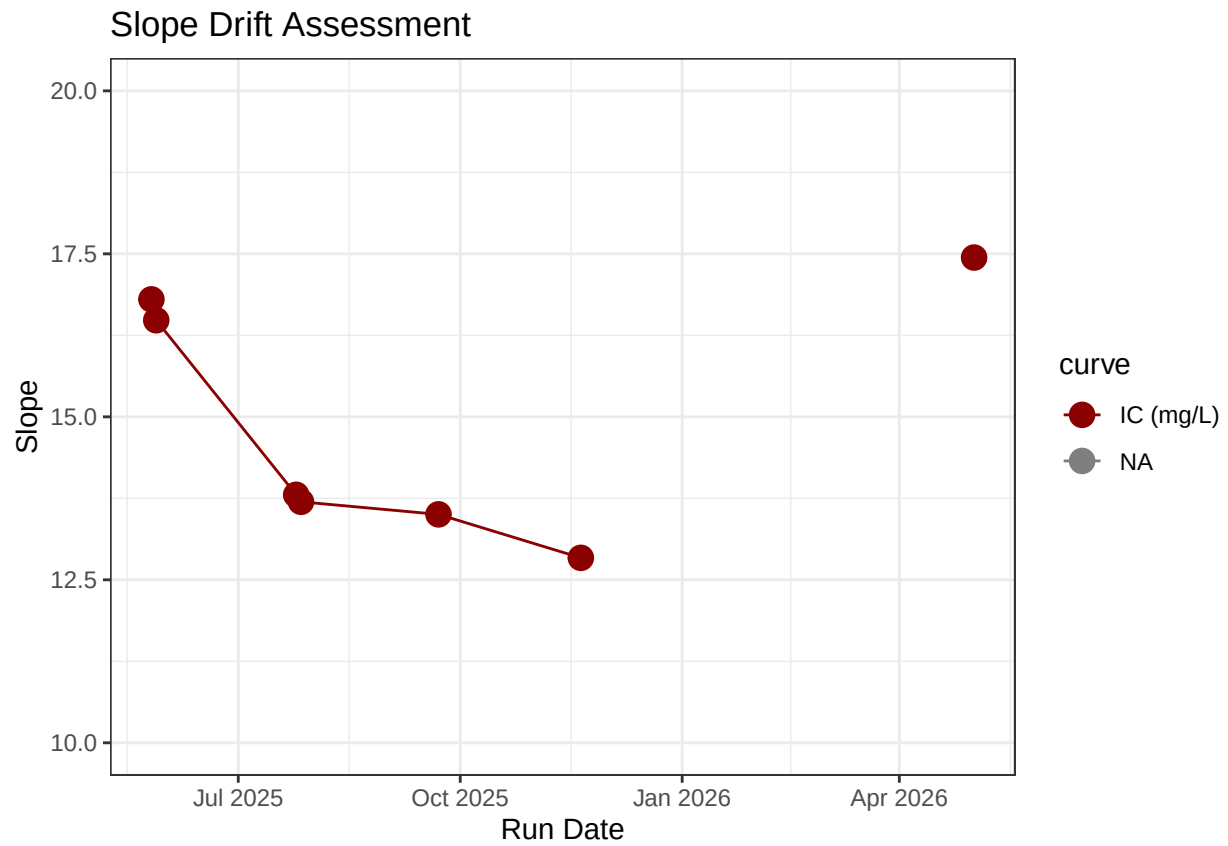
```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_line()').
```

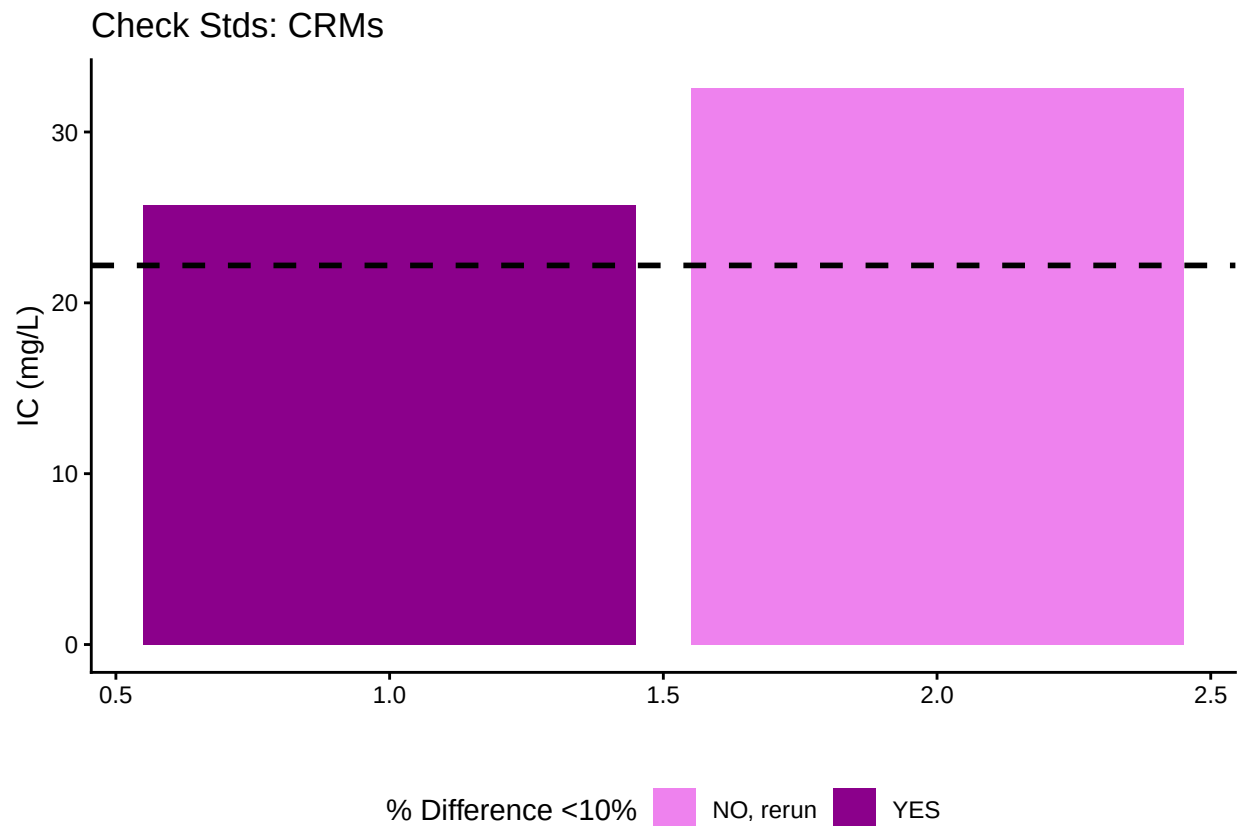


```
## [1] "IC Curve r2 GOOD"
```

0.4 CRMs Check: CRM Batch 221-0866

```
## Assess the CRMs
```

```
## New names:
## * ' ' -> '...14'
```



```
## [1] "IC crm has a % Difference >25% of expected - REASSESS"
```

```
## Run mean = 29.16
```

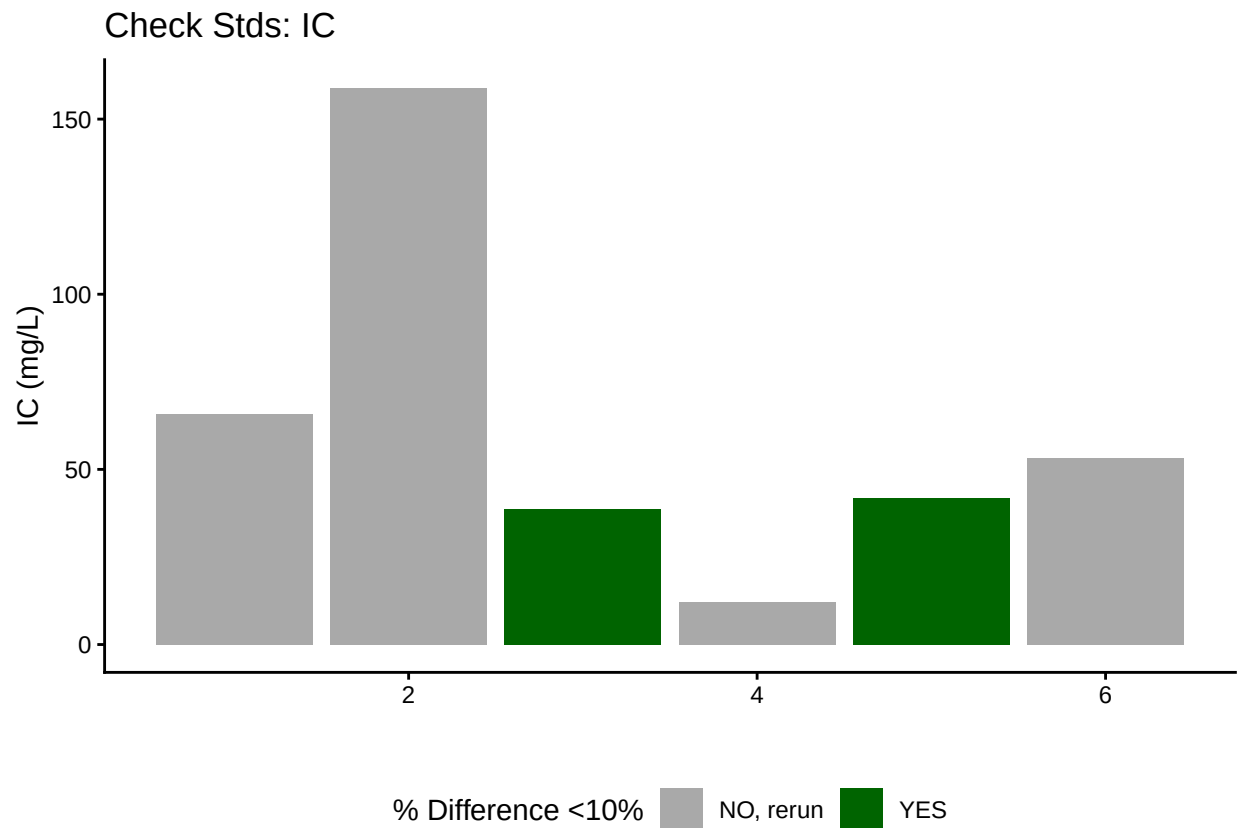
```
## Expected = 22.19
```

0.5 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
```

```
## * ' ' -> '...14'
```



```
## [1] ">60% of IC Check Standards are within range of expected concentration"
```

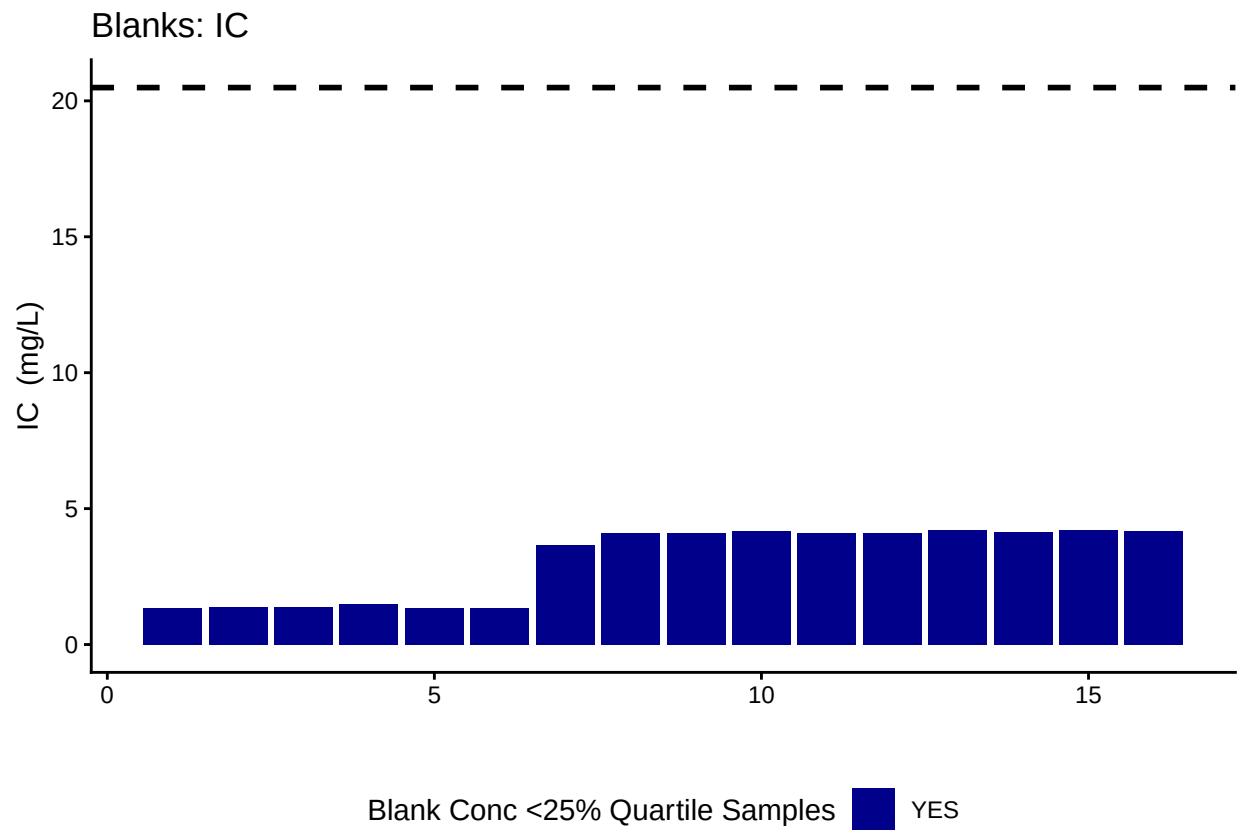
0.6 Assess Blanks

```
## Assess Blanks
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

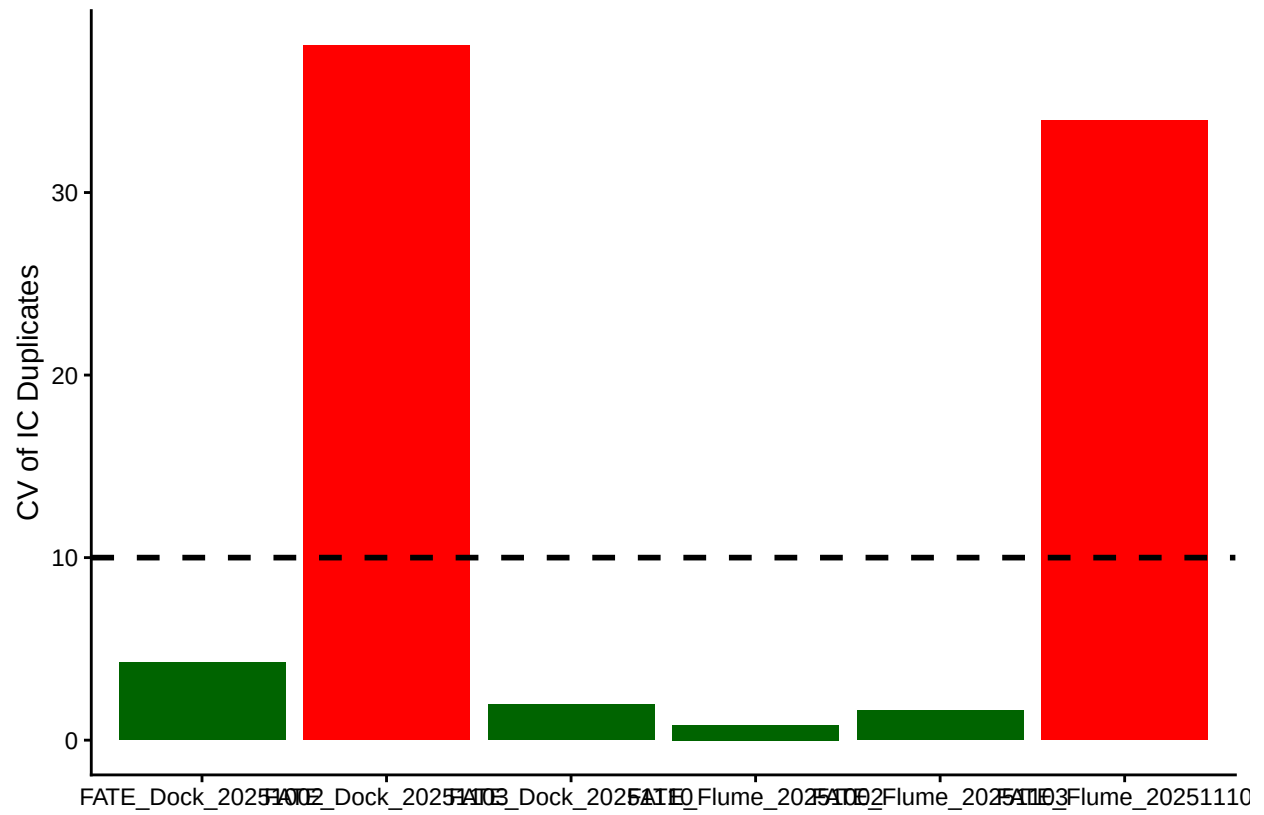


```
## carbon blanks:
```

```
## [1] 3.078562
```

0.7 Assess Duplicate Samples

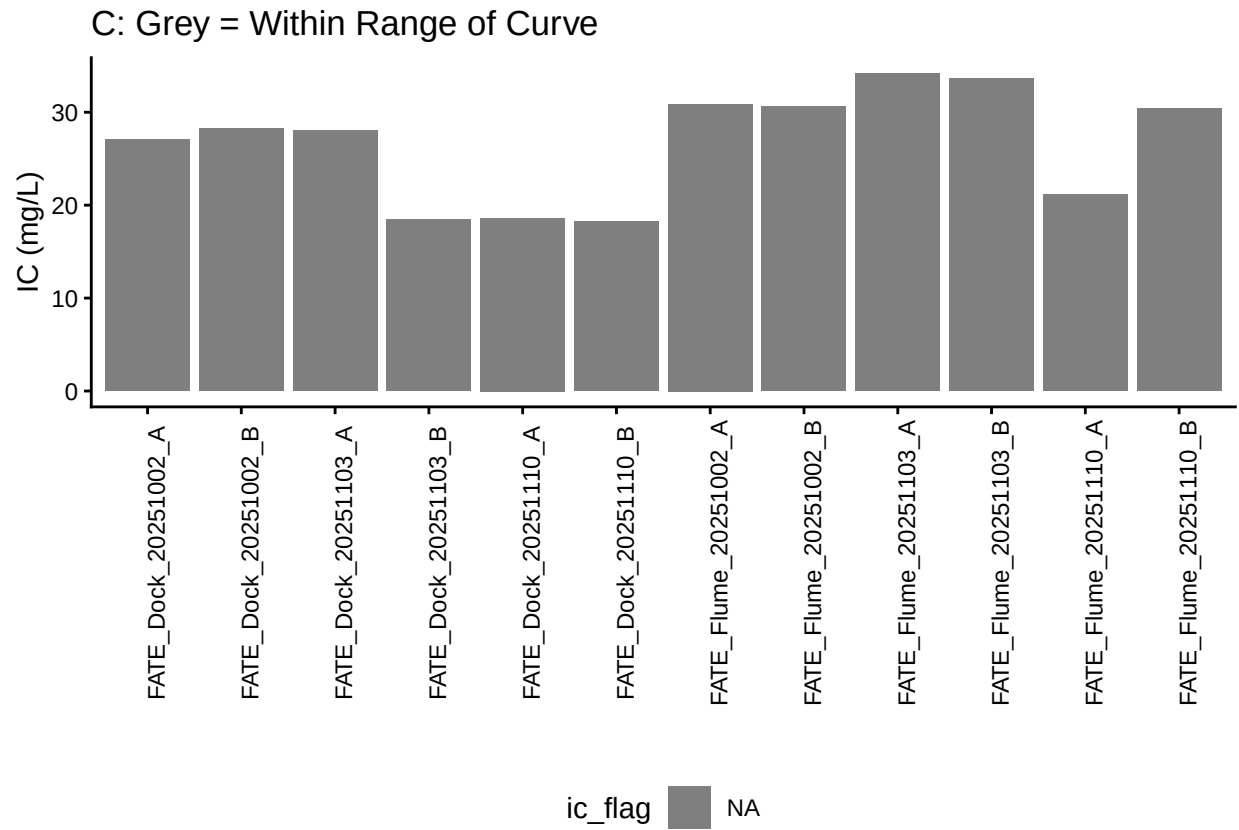
```
## Assess Duplicates
```



```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

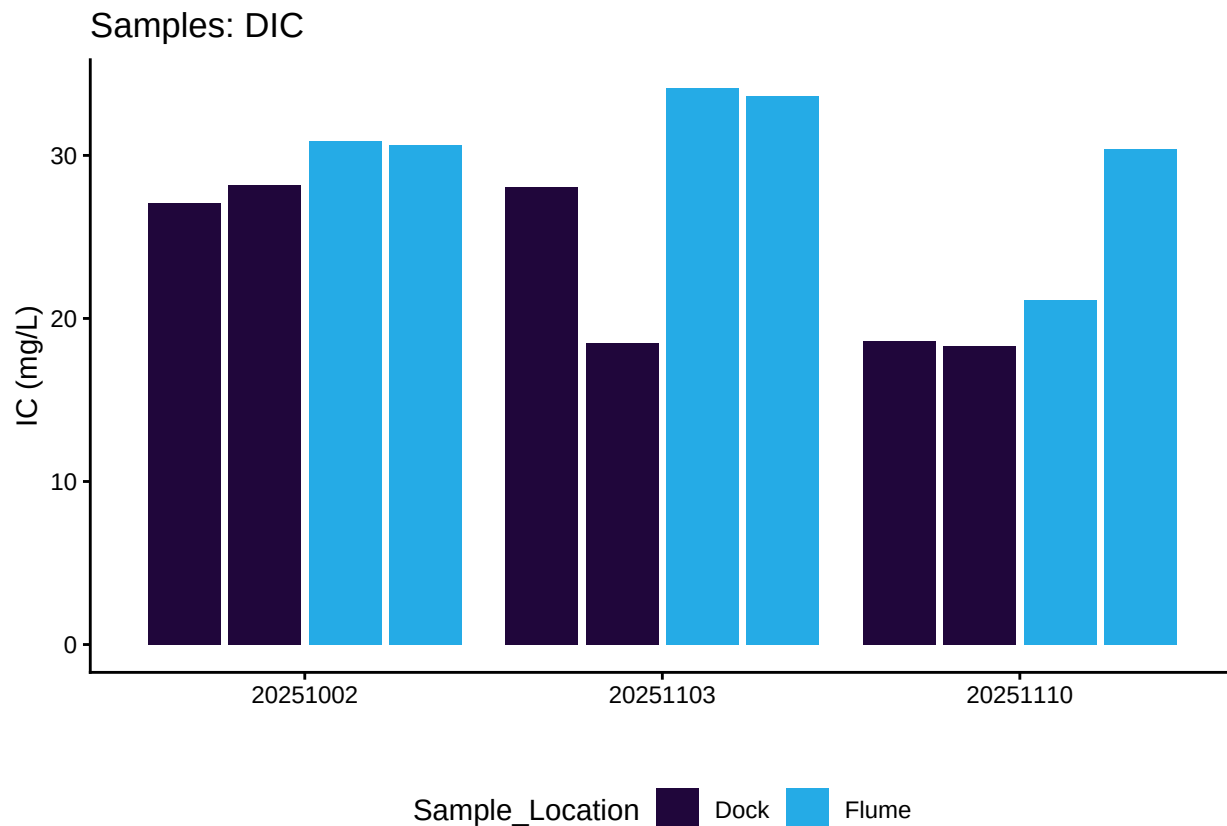
0.8 Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

0.9 Visualize Data by Dock & Flume & Date

Visualize Data



0.10 Convert data from mg/L to uMoles/L

0.11 Export Processed Data

Export Processed Data

```
## # A tibble: 6 x 9
##   Project Sample_Location Replicate Date   ic_mgL ic_uM ic_flag Analysis_runtime
##   <chr>    <chr>           <chr>   <chr>   <dbl> <dbl> <chr>   <chr>
## 1 FATE    Dock             A      20251~  27.1  2255. IC che~ 11/20/2025 11:0~
## 2 FATE    Dock             B      20251~  28.2  2348. IC che~ 11/20/2025 11:2~
## 3 FATE    Flume            A      20251~  30.9  2572. IC che~ 11/20/2025 11:5~
## 4 FATE    Flume            B      20251~  30.6  2552. IC che~ 11/21/2025 12:1~
## 5 FATE    Dock             A      20251~  28.0  2335. IC che~ 11/21/2025 12:3~
## 6 FATE    Dock             B      20251~  18.5  1539. IC che~ 11/21/2025 1:00~
## # i 1 more variable: Run_notes <chr>
```

#end